# CAST Science Data Analysis: CER Sentence Starters & Formula Guide

NGSS SEP-4 & SEP-7 • Reference Sheet

**UNIT 2: 2D PROJECTILE MOTION • STUDENT DESK REFERENCE & WRITING TOOLKIT**

## 1. PROJECTILE MOTION VOCABULARY & FORMULA TOOLKIT (g = 10 m/s²)

Use These Terms in Your Reasoning

**Horizontal Speed ($v_{ox}$)** | **Initial Vertical Speed ($v_{oy}$)** | **Hang Time ($t_{hang}$)** | **Landing Range ($\Delta x$)**

**Linear Trend (Constant Slope)** | **Complementary Angles (Add to 90°)** | **Optimal Angle (45°)**

- **Horizontal Range Formula:** $\Delta x = v_{ox} \times t$
- **Slope of Range vs. Speed Graph:** Slope = $\Delta x \div v_{ox}$ = Flight Time ($t$)
- **Level-Ground Hang Time:** $t_{hang} = 2v_{oy} \div g$

## 2. ZONE A — CLAIM SENTENCE STARTERS (1 Point)

Directly Answer the Question & State the Overall Pattern

**Option 1 (Evaluating a Student Misconception):**
"The student's claim is **incorrect** because increasing the launch angle past **45°** causes the horizontal landing range ($\Delta x$) to **decrease**, not increase."

**Option 2 (Comparing Complementary Angles like 30° and 60°):**
"The student's argument is **incorrect** because launch angles of **30°** and **60°** are complementary angles (adding to 90°) that produce the **exact same horizontal landing range of** _____ **m** on level ground."

**Option 3 (Describing Graph Trends):**
"As **[ independent variable ]** increases from _____ to _____, the **[ dependent variable ]** increases linearly / reaches a maximum at **45°** and then decreases symmetrically."

## 3. ZONE B — QUANTITATIVE EVIDENCE SENTENCE STARTERS (2 Points)

Cite At Least TWO Data Points + Units!

**Comparing Peak Angle (45°) vs. Steep Angles (60° or 75°):**
• "According to **Table 2 and Graph 2**, at a launch angle of **45°**, the hang time is **2.8 s** and the projectile reaches its maximum landing range of **40 m**."
• "In contrast, when the launch angle increases to **60°** (or **75°**), the hang time increases to **3.5 s** (or **3.9 s**), but the landing range drops to **35 m** (or **20 m**)."

**Comparing Complementary Angles (30° vs. 60°):**
• "In **Table 2**, a **30°** launch has a fast horizontal speed ($v_{ox}$ = **17.3 m/s**) and a shorter hang time ($t_{hang}$ = **2.00 s**), landing at **34.6 m**."
• "Meanwhile, a **60°** launch has a slower horizontal speed ($v_{ox}$ = **10.0 m/s**) and a longer hang time ($t_{hang}$ = **3.46 s**), landing at the exact same range of **34.6 m**."

## 4. ZONE C — PHYSICS REASONING SENTENCE STARTERS (1 Point)

Explain WHY Using the Equation $\Delta x = v_{ox} \times t_{hang}$

**Explaining the Trade-Off Between Horizontal Speed and Hang Time:**
"Because horizontal landing range is determined by multiplying **Horizontal Speed × Hang Time ($\Delta x = v_{ox} \times t_{hang}$)**, launching at a very steep angle gives the ball more vertical hang time, but leaves it with too little horizontal speed ($v_{ox}$) to travel far forward."

**Explaining Why 30° & 60° Tie and Why 45° Wins:**
"For complementary angles like **30° and 60°**, the horizontal and vertical velocity components trade places, so the product $v_{ox} \times t_{hang}$ equals **34.6 m** for both. A **45°** launch maximizes range (**40.0 m**) because it splits launch speed equally between horizontal speed and vertical hang time."